CHE 321 - Exam 2 - Sample 2

examProblem_fun.m

```
function h=examProblem_fun(v,t)
Fa = t(1);
Fb = t(2);
Fc = t(3);
Fd = t(4);
T = t(5);
Ta = t(6);

Cpa = 10;
Cpb = 10;
Cpc = 30;
Cpd = 20;

E2 = 12000;
y = 1;
R = 1.9872;
E1 = 8000;
T0 = 300;
Ct0 = 0.2;
Cpco = 10;
m = 50;
DH1b = -15000;
DH2a = -10000;
kc = 0.5;
Ua = 80;

Ft = Fa+Fb+Fc+Fd;

k1a = 40*exp((E1/R)*(1/300-1/T));
K2c = 2*exp(((E2/R*(1/300-1/T))));
sumFiCPI = Cpa*Fa+Cpb*Fb+Cpc*Fc+Cpd*Fd;

Ca = Ct0*(Fa/Ft)*(T0/T)*y;
Cb = Ct0*(Fb/Ft)*(T0/T)*y;
Cc = Ct0*(Fc/Ft)*(T0/T)*y;

r2c = -K2c*Ca*Cc;
r1a = -k1a*Ca*Cb^2;
r1b = 2*r1a;
rb  = r1b;
r2a = r2c;

r1c = -r1a;
rc = r1c + r2c;
r2d =-2*r2c;
ra = r1a+r2a;
rd = r2d;

Qg = r1b*DH1b+r2a*DH2a;
Qr = Ua*(T-Ta);

h1=ra;
h2=rb;
h3=rc-kc*Cc;
h4=rd;
h5=(Qg-Qr)/sumFiCPI;
h6=Ua*(Ta-T)/m/Cpco;

h = [h1 h2 h3 h4 h5 h6]';
```

examProblem_simi.m

```
clc
clear all;

t0 = [5 10 0 0 300 511.9855];

vspan = [0 10];

[v,t] = ode45(@examProblem_fun,vspan,t0);

Fa = t(:,1);
Fb = t(:,2);
Fc = t(:,3);
Fd = t(:,4);
T = t(:,5);

Ta = t(:,6);

figure(1)
plot(v,Fa)
hold on
plot(v,Fb)
plot(v,Fc)
plot(v,Fd)
legend('Fa','Fb','Fc','Fd')
figure(2)
plot(v,T)
hold on
plot(v,Ta)
legend('T','Ta')

T0=300;
y=1;
Ft=Fa+Fb+Fc+Fd;
Ct0=0.2;

Ca=Ct0.*(Fa./Ft).*(T0./T).*y;
Cb=Ct0.*(Fb./Ft).*(T0./T).*y;
Cc=Ct0.*(Fc./Ft).*(T0./T).*y;
Cd=Ct0.*(Fd./Ft).*(T0./T).*y;

figure(3)
plot(v,Ca)
hold on
plot(v,Cb)
plot(v,Cc)
plot(v,Cd)
legend('Ca','Cb','Cc','Cd')
```